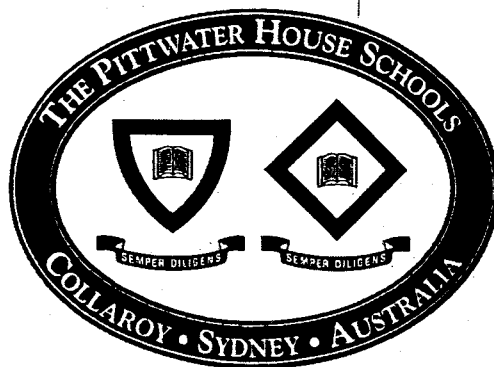


NAME: _____

STUDENT No: _____



YEAR 12 TRIAL HSC EXAMINATION 2004

MATHEMATICS

*Time Allowed: 3 hours
(plus 5 minutes reading time)*

INSTRUCTIONS FOR CANDIDATES

- Start each question in a new booklet.
- All questions are to be answered.
- All questions are of equal value.
- The marking scheme has been given on the right hand side of the page.
- Standard integrals are printed on the back page.
- Approved calculators may be used.

Question 1.
(Start a new booklet)

Marks

- (a) $|3 - x| = 7$ 2
- (b) Differentiate $y = 2\sqrt{x} + \cos x$ with respect to x . 2
- (c) An arc of length 8 cm subtends an angle of θ at the centre of a circle with radius 3 cm. Find θ to the nearest degree. 2
- (d) Differentiate $f(x) = \ln\left(\frac{x+1}{x-1}\right)$. Answer in simplest form. 2
- (e) Evaluate $e^{-2.6}$ correct to 3 significant figures. 2
- (f) At a sale, 30% discount is given off the marked price. If Anna pays \$175 for a coat at this sale, what was the original marked price? 2

Question
(Start a

(a) Fi

(b) Li

Li

(i)

(ii)

(iii)

(iv)

(v)

(vi)

(vii)

Question 2.
(Start a new booklet)

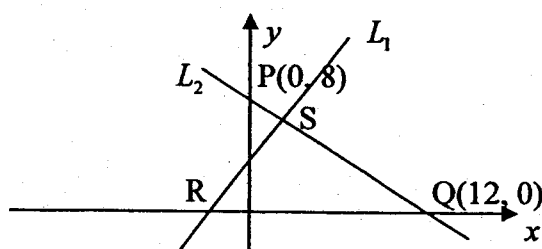
Marks

- (a) Find the equation of the normal to $y = 2e^{2x}$ at the point $x = 0$

3

- (b) Line L_1 has the equation $3x - 2y + 3 = 0$

Line L_2 passes through $P(0, 8)$ and $Q(12, 0)$



- (i) L_1 cuts the x axis at R . Find the coordinates of R .
- (ii) Find the gradient of line L_2
- (iii) Show L_1 and L_2 are perpendicular
- (iv) Show the equation of L_2 is $2x + 3y - 24 = 0$
- (v) Find S , the point of intersection of L_1 and L_2
- (vi) Find the area of ΔRSQ
- (vii) Copy the diagram and shade the region where $2x + 3y - 24 \leq 0$, $3x - 2y + 3 \geq 0$ and $y \geq 0$ hold simultaneously.

Question 3.
(Start a new booklet)

Marks

Question
(Start a

(a) Find $\int \frac{2x}{x^2 - 4} dx$

1

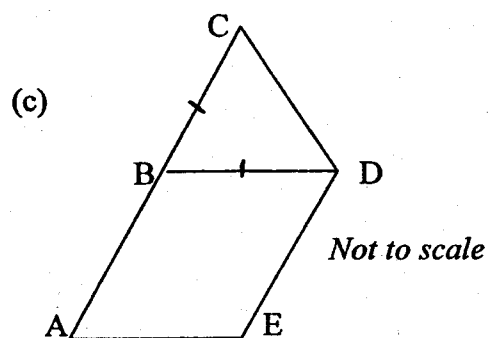
(a) A p
of 2

At
trav
tow

(b) Evaluate $\int_0^{\frac{\pi}{2}} 1 + \cos x dx$

2

(i)



A, B and C are collinear

ABDE is a parallelogram

$\triangle BCD$ is isosceles with $BC = BD$

$\angle BCD = 58^\circ$

(ii)

(b) A p

(i)

(ii)

(i) Copy the diagram

(ii) Find the size of $\angle AED$ giving reasons

3

(iii)

(iv)

(d) Differentiate with respect to x

2

(v)

(i) $(3x^2 - 5)^7$

(ii) $x^3 \tan x$

(c) The c

(i)

(ii)

(e) $\int_0^1 (3x+1)^3 dx$

2

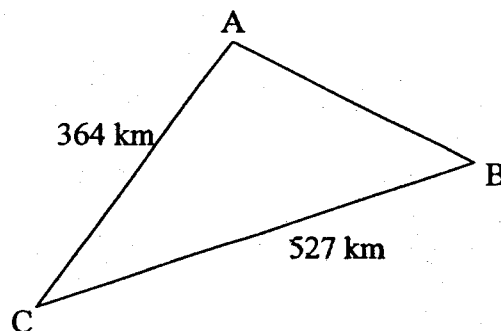
Question 4.

(Start a new booklet)

Marks

- (a) A plane is flying from town A on a bearing of 237°T to town C which is 364 km away.

At town C, the plane changes its course and travels 527 km on a bearing of 077°T to town B.



- (i) Copy the diagram and show $\angle ACB = 20^\circ$

2

- (ii) Find the distance of B from A. Answer to the nearest km.

2

- (b) A parabola has the point (2, 1) as its focus and its directrix is $y = -3$. Write down:

5

- (i) the vertex

- (ii) the equation of its axis

- (iii) the focal length

- (iv) the equation of the parabola; and

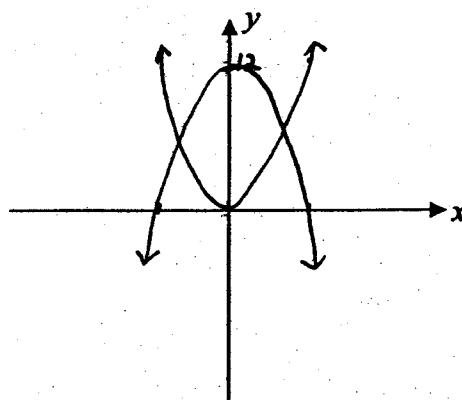
- (v) draw its sketch.

- (c) The curves $y = 2x^2$ and $y = 12 - x^2$ are drawn

3

- (i) Show the curves intersect at (-2, 8) and (2, 8)

- (ii) Hence, find the area between the curves



Question 5.
(Start a new booklet)

Marks

- (a) The first 3 terms of a geometric series are $x - 2$, $x + 1$, $3x - 3$.
- (i) Find the 2 values of x
- (ii) Write down the 2 series and find their ratios
- (iii) Explain why these series don't have a limiting sum
- (b) The gradient function of a curve is given by $3x^2 - 11$. If the curve passes through (3, 4) find its equation.
- (c) Find k if $4x^2 - kx + 1 = 0$ has real roots.
- (d) Solve $2\sin x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$

2

1

1

3

3

2

Question 6.
(Start a new booklet)

(a) Solve the equation

(b) A curve has equation $y = x^2 + 2x + 3$. Find the coordinates of the vertex of the curve.

Determine the set of values of x for which the curve is above the x -axis.

Consider the function $f(x) = x^2 + 2x + 3$. Find the range of $f(x)$ for $0 \leq x \leq 4$.

(i) Find the maximum value of $f(x)$.

(ii) Find the minimum value of $f(x)$.

(iii) Find the values of x for which $f(x) = 5$.

(iv) Find the values of x for which $f(x) = 7$.

(c) A particle moves in a straight line from rest with constant acceleration. It travels a distance of 100 m in 10 s. Find the acceleration.

(i) Find the acceleration.

(ii) Find the speed of the particle at the end of 10 s.

(d) A car starts from rest and accelerates uniformly to a speed of 30 m/s in 10 s. It then travels at this speed for 20 s. Find the total distance travelled by the car.

Question 6.
(Start a new booklet)

Marks

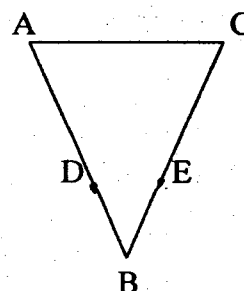
(a) Solve $\log_3(2x - 4) = 4$

2

(b) ABC is a triangle with $AB = BC$.

D and E are points of AB and CB such that $AD = CE$.

Copy the diagram.



(i) Why does $DB = EB$?

1

(ii) Prove $\triangle ABE$ is congruent to $\triangle CBD$.

2

(iii) Why does $\angle BAE = \angle BCD$?

1

(iv) Prove $\angle CAE = \angle ACD$.

1

(c) A point P (x, y) moves so that its distance from A(3, 0) is twice its distance from B(0, 3).

(i) Show the equation of the locus of P is $x^2 + 2x + y^2 - 8y + 9 = 0$

2

(ii) Show this locus is a circle and write down its centre and radius.

2

(d) Change $\frac{3\pi}{5}$ radians to degrees.

1

Question 7.
(Start a new booklet)

Marks

(a) For the curve $y = \frac{1}{3}x^3 - x^2 - 8x + 12$,

- (i) Find any turning points.
- (ii) Determine their nature
- (iii) Find any points of inflexion
- (iv) Sketch the curve showing all relevant features
- (v) For what values of x is the curve concave up?
- (vi) When is the curve decreasing?

2

1

2

1

1

1

(b) An arithmetic progression has a first term of 1 and a last term of 14. If the sum of the series is 90,

- (i) Find the number of terms in the series
- (ii) Show the difference is $\frac{13}{11}$
- (iii) Find an expression for the n th term as a single fraction.

2

1

1

Question
(Start a new booklet)

(a) Find

(b) Es

Ar

(c) So

(d) Wh
The
at t

(i)

(ii)

(iii)

Question 8.

(Start a new booklet)

Marks

- (a) Find A , B and C if $3x^2 + 4x + 2 \equiv Ax(x-1) + B(x+1) + C$ 3
- (b) Estimate $\int_0^1 \sin(1+x^2) dx$ using Simpson's Rule with 3 function values.
Answer to three decimal places. 3
- (c) Solve $2\sin^2 x - 3\sin x + 1 = 0$ for $0 \leq x \leq 2\pi$ 2
- (d) When Tom started university, his parents borrowed \$30000 to pay for his education. They repaid the loan by making 48 equal monthly repayments. Interest was charged at the rate of 1.1% per month on balance owing.
- (i) Write an expression for the amount owing after 1 monthly repayment of \$ m . 1
- (ii) Show that after 2 monthly repayments, the amount owing is $\$30663.63 - 2.011m$. 1
- (iii) Calculate the value of each monthly repayment. 2
Answer to the nearest cent.

Question 9.

(Start a new booklet)

Marks

(a) If α and β are roots of $3x^2 + 5x - 2 = 0$, find:

(i) $\alpha + \beta$

1

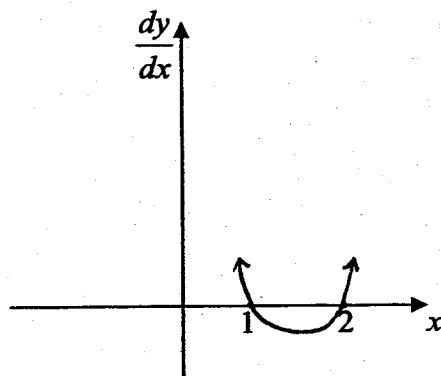
(ii) $\alpha\beta$

1

(iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$

2

(b) The sketch shows the graph of $\frac{dy}{dx}$ which is the gradient function of $y = f(x)$.

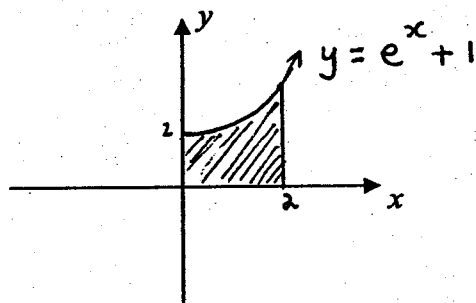


If $f(0) = 1$, draw a possible sketch of $y = f(x)$.

2

(c) What is the volume of the solid formed when the shaded area is rotated about the x axis?

3



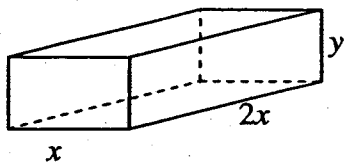
(d) Solve $\frac{1}{2}(5x-1) = 4 - \frac{1}{3}(x-2)$

3

Question 10.
(Start a new booklet)

Marks

- (a) Boxes in the shape of rectangular prisms are to be made so that the width (x metres) of the base is half the length.



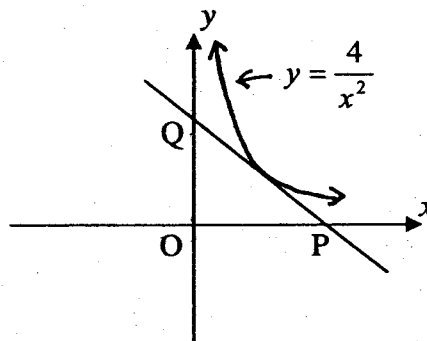
Each box holds a volume of 4 cubic metres. The wood used to make the top and bottom of the box costs \$15 per m^2 while the wood for the other 4 sides costs \$10 per m^2 .

- (i) Find an expression for y in terms of x . 1
- (ii) Show the cost \$ C of building each box is given by $C = 60x^2 + \frac{120}{x}$. 2
- (iii) What are the measurements of the cheapest boxes that can be constructed? 3

- (b) PQ is a tangent to $y = \frac{4}{x^2}$.

It cuts the x axis at P and the y axis at Q.

$$\angle OPQ = \angle OQP$$



- (i) Explain why the gradient of PQ is -1. 1
- (ii) Show PQ is a tangent touching the curve at the point (2, 1). 2
- (iii) Find the equation of this tangent. 1
- (c) Differentiate $y = \cot x$. Answer in the simplest form. 2

Question 1

$$|3-x| = 7$$

$$3-x = 7 \quad \text{or} \quad 3-x = -7$$

$$x = -4 \quad \text{or} \quad x = 10$$

$$y = 2\sqrt{x} + \cos x = 2x^{\frac{1}{2}} + \cos x$$

$$\frac{dy}{dx} = x^{-\frac{1}{2}} - \sin x$$

$$= \frac{1}{\sqrt{x}} - \sin x$$

$$l = 8 \quad r = 3$$

$$l = 10 \quad \therefore 8 = 30$$

$$\theta = \frac{8}{3} \text{ radians}$$

$$= \frac{8}{3} \times \frac{180}{\pi}$$

$$= 152^{\circ} 47'$$

$$= 153^{\circ} \text{ (to nearest deg.)}$$

$$f(x) = \ln\left(\frac{x+1}{x-1}\right) = \ln(x+1) - \ln(x-1)$$

$$f'(x) = \frac{1}{x+1} - \frac{1}{x-1} = \frac{x-1-(x+1)}{(x+1)(x-1)}$$

$$= \frac{x-1-x-1}{x^2-1} = \frac{-2}{x^2-1}$$

$$-2.6$$

$$= 0.074273578$$

$$= 0.0743 \text{ to 3 sig. figs.}$$

$$70\% = \$175$$

$$1\% = \frac{175}{70} = 2.5$$

$$100\% = 250$$

$$\therefore \text{Original price} = \$250$$

Question 2

$$y = 2e^{2x}$$

$$y = 2e^{2x} \times 2 = 4e^{2x}$$

$$\text{At } x=0, \frac{dy}{dx} = 4 \quad y = 2$$

$$\therefore m=4 \quad P=(0,2) \quad m_{\text{normal}} = -\frac{1}{4}$$

Eqn of normal is

$$y-2 = -\frac{1}{4}(x-0)$$

$$4y-8 = -x$$

$$x+4y-8=0$$

i) L_1 is $3x-2y+3=0$

Cuts x axis when

$$y=0 \quad \text{ie } 3x+3=0$$

$$x=-1$$

$$\therefore R = (-1,0)$$

$$ii) L_2 \text{ has } m = \frac{8-0}{0-12} = -\frac{2}{3}$$

$$iii) 3x-2y+3=0$$

$$2y = 3x+3$$

$$y = \frac{3}{2}x + \frac{3}{2}$$

$$m = \frac{3}{2}$$

$$m_{L_1} \times m_{L_2} = \frac{3}{2} \times -\frac{2}{3} = -1$$

$\therefore L_1$ and L_2 are $\perp$

iv) Equation of L_2 is

$$y-8 = -\frac{2}{3}(x-0)$$

$$3y-24 = -2x$$

$$2x+3y-24=0$$

$$v) 3x-2y=-3 \quad (1)$$

$$2x+3y=24 \quad (2)$$

$$\times (1) \text{ by } 3$$

$$9x-6y=-9 \quad (3)$$

$$\times (2) \text{ by } 2$$

$$4x+6y=48 \quad (4)$$

$$(3)+(4)$$

$$13x = 39$$

$$x = 3, y = 6$$

$$\therefore S = (3,6)$$

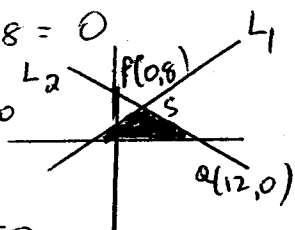
$$vi) R = (-1,0) \quad Q = (12,0)$$

$$\therefore RQ = 13 \quad \text{Height is } 6$$

$$\text{Area} = \frac{1}{2} \times 13 \times 6$$

$$= 39 \text{ u}^2$$

vii) See diagram.



Question 3

$$\int \frac{2x}{x^2-4} dx = \ln(x^2-4) + C$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} 1 + \cos x dx &= \left[x + \sin x \right]_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{2} + \sin \frac{\pi}{2} - (0 + \sin 0) \\ &= \frac{\pi}{2} + 1 \end{aligned}$$

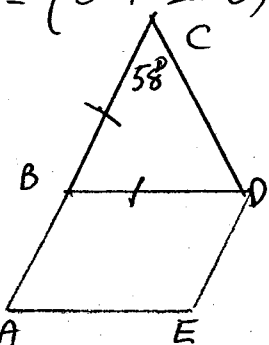
i) 1)

$$\text{ii) } \angle BDC = \angle BCD = 58^\circ$$

Base L's of isos. Δ) A

$$\angle ABD = \angle BDC + \angle BCD = 58^\circ + 58^\circ = 116^\circ$$

$$\angle AED = 116^\circ (\text{opp. L's of parm. } =)$$



$$\begin{aligned} \text{1) i) } \frac{d}{dx} (3x^2-5)^7 &= 7(3x^2-5)^6 \times 6x \\ &= 42x(3x^2-5)^6 \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{d}{dx} x^3 + \tan x &= x^3 \times \sec^2 x + \tan x \times 3x^2 \\ \text{or } &= x^2(x \sec^2 x + 3 \tan x) \end{aligned}$$

$$\begin{aligned} \int_0^1 (3x+1)^3 dx &= \left[\frac{(3x+1)^4}{4 \times 3} \right]_0^1 \\ &= \frac{256}{12} - \frac{1}{12} \\ &= \frac{255}{12} \text{ or } 21\frac{1}{4} \end{aligned}$$

Question 4

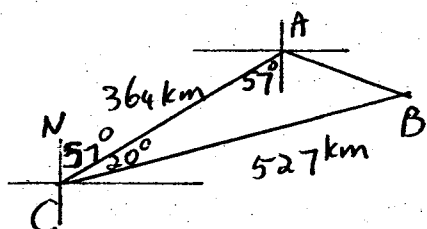
$$\text{i) } \angle NCA = 57^\circ$$

(alt L's =)

$$\angle NCB = 77^\circ$$

$$= 57 + 20$$

$$\therefore \angle ACB = 20^\circ$$



$$\begin{aligned} \text{ii) } AB^2 &= 364^2 + 527^2 - 2 \times 364 \times 527 \times \cos 20^\circ \\ &= 49706.28788 \end{aligned}$$

$$AB = 222.949$$

$$= 223 \text{ km to nearest km}$$

then x

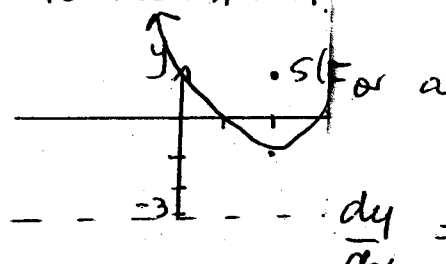
$$\text{b) i) } V = (2, -1)$$

$$\text{ii) axis is } x = 2$$

$$\text{iii) } a = 2$$

$$\text{iv) } (x-2)^2 = 8(y+1)$$

v) See sketch



$$\begin{aligned} \text{c) i) } y &= 2x^2 \text{ and } y = 12 - x^2 \\ \text{intersect when } 2x^2 &= 12 - x^2 \end{aligned}$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = 2$$

$$\text{or } x = -2$$

$$y = 8$$

$$y = 8$$

$$\begin{aligned} \text{ii) Area} &= 2 \int_0^2 (12 - x^2 - 2x^2) dx \\ &= 2 \int_0^2 (12 - 3x^2) dx \\ &= 2 \left[12x - x^3 \right]_0^2 \\ &= 2(24 - 8) \\ &= 32 \text{ u}^2 \end{aligned}$$

Question 5

$$\text{a) } x-2, x+1, 3x-3$$

$$\text{i) } \frac{x+1}{x-2} = \frac{3x-3}{x+1}$$

$$x^2 + 2x + 1 = 3x^2 - 9x + 6$$

$$2x^2 - 11x + 5 = 0$$

$$(2x-1)(x-5) = 0$$

$$x = \frac{1}{2} \text{ or } 5$$

Question 6

$$9_3 (2x)$$

$$2x$$

$$2$$

(3) When $x = \frac{1}{2}$, Series is $-\frac{3}{2}, \frac{3}{2}, -\frac{3}{2}$
ratio is -1

When $x = 5$, Series is $3, 6, 12$
ratio is 2

For a limiting sum, $|r| < 1$.

$$\frac{dy}{dx} = 3x^2 - 11$$

$$y = x^3 - 11x + C$$

Curve passes thru $(3, 4)$

$$\therefore 4 = 27 - 33 + C$$

$$C = 10$$

$$\text{Curve is } y = x^3 - 11x + 10$$

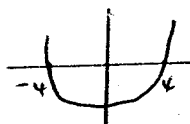
$4x^2 - kx + 1 = 0$ has real

$$\text{roots } \therefore \Delta \geq 0$$

$$k^2 - 16 \geq 0$$

$$(k+4)(k-4) \geq 0$$

$$k \leq -4 \text{ or } k \geq 4$$



$$2 \sin x = -\sqrt{3} \text{ for } 0 \leq x \leq 2\pi$$

$$\sin x = -\frac{\sqrt{3}}{2}$$

x is in quadrants 3 or 4

$$x = \pi + \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$$

$$= \frac{4\pi}{3} \text{ or } \frac{5\pi}{3}$$

Q. 6

$$9(2x-4) = 4$$

$$2x-4 = \frac{4}{9}$$

$$2x-4 = 81$$

$$2x = 85$$

$$x = 42.5$$

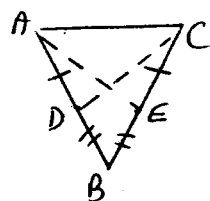
$$b) i) AB = AD + DB$$

$$CB = CE + EB$$

$$\text{But } AB = CB$$

$$\therefore AD + DB = CE + EB$$

$$\text{But } AD = CE, \therefore DB = EB$$



$$ii) AB = BC \text{ given.}$$

$$EB = DB \text{ (proven above)}$$

$\angle B$ is common.

$$\therefore \triangle ABE \equiv \triangle CBD \text{ (SAS)}$$

$$iii) \angle BAE = \angle BCD \text{ (corresp. } \angle \text{'s of } \triangle \text{'s)}$$

$$iv) \angle BAC = \angle BCA \text{ (base } \angle \text{'s of } \triangle \text{'s)}$$

$$\angle BAE + \angle CAE = \angle BCD + \angle ACD$$

$$\text{But } \angle BAE = \angle BCD \text{ (proven above)}$$

$$\therefore \angle CAE = \angle ACD$$

$$c) P(x, y) \quad A(3, 0) \quad B(0, 3)$$

$$i) PA = 2PB$$

$$\therefore (PA)^2 = 4(PB)^2$$

$$(x-3)^2 + y^2 = 4[x^2 + (y-3)^2]$$

$$x^2 - 6x + 9 + y^2 = 4x^2 + 4y^2 - 24y + 36$$

$$3x^2 + 3y^2 + 6x - 24y + 27 = 0$$

$$x^2 + y^2 + 2x - 8y + 9 = 0$$

$$ii) x^2 + 2x + 1 + y^2 - 8y + 16 = -9 + 16$$

$$(x+1)^2 + (y-4)^2 = 8$$

This is a circle centre $(-1, 4)$

$$\text{radius} = 2\sqrt{2}$$

$$d) \frac{3\pi}{5} = \frac{3 \times 180}{5} = 108^\circ$$

Question 7

$$y = \frac{1}{3}x^3 - x^2 - 8x + 12$$

$$\frac{dy}{dx} = x^2 - 2x - 8$$

For turning pt, $\frac{dy}{dx} = 0$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$x = 4 \quad \text{or} \quad x = -2$$

$$y = -14\frac{2}{3} \quad \text{or} \quad y = 21\frac{1}{3}$$

$$\frac{d^2y}{dx^2} = 2x - 2$$

$$\text{When } x = 4, \frac{d^2y}{dx^2} > 0$$

$\therefore$ min. at $(4, -14\frac{2}{3})$

$$\text{When } x = -2, \frac{d^2y}{dx^2} < 0$$

$\therefore$ max. at $(-2, 21\frac{1}{3})$

i) For a pt. of inflexion,

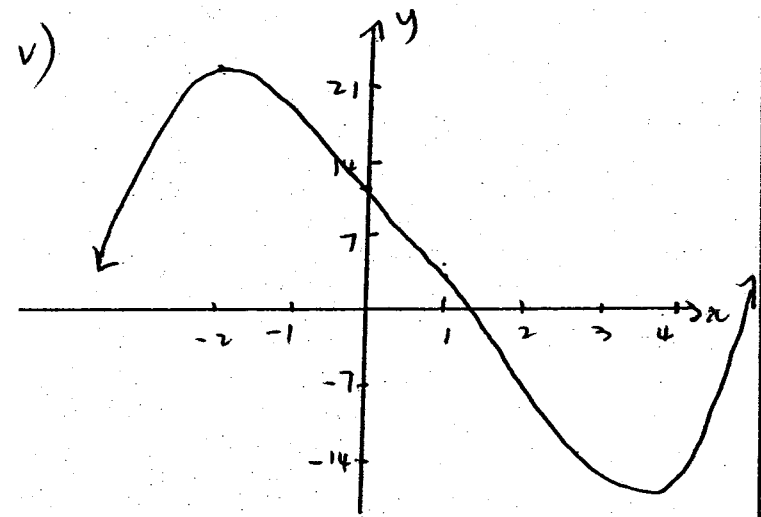
$$\frac{d^2y}{dx^2} = 0 \quad \text{ie} \quad 2x - 2 = 0$$

$$x = 1$$

$$y = 3\frac{1}{3}$$

When $x < 1$ $\frac{d^2y}{dx^2} < 0$ } Change in
 $x > 1$ $\frac{d^2y}{dx^2} > 0$ } Concavity

$\therefore$ Pt of inflexion at $(1, 3\frac{1}{3})$

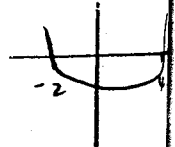


v) Curve is concave up
 for $\frac{d^2y}{dx^2} > 0$ ie $2x - 2 > 0$
 $x > 1$

vi) Curve is decreasing when

$$\frac{dy}{dx} < 0 \quad \text{ie} \quad (x-4)(x+2) < 0$$

$$-2 < x < 4$$



b) 1, ..., 14

$$90 = \frac{n}{2}(1 + 14)$$

$$180 = 15n$$

$$n = 12 \quad \therefore 12 \text{ terms}$$

$$ii) 14 = 1 + 11d$$

$$13 = 11d$$

$$d = \frac{13}{11}$$

$$iii) t_n = a + (n-1)d$$

$$= 1 + (n-1)\frac{13}{11}$$

$$= \frac{11 + 13n - 13}{11}$$

$$= \frac{13n - 2}{11}$$

Question 8

$$a) 3x^2 + 4x + 2 \equiv Ax(x-1) + B(x+1)$$

$$3x^2 + 4x + 2 \equiv Ax^2 - Ax + Bx + B$$

$$\therefore 3 = A$$

$$4 = -A + B$$

$$4 = -3 + B$$

$$B = 7$$

$$2 = B + C$$

$$2 = 7 + C$$

$$C = -5$$

$$\therefore A = 3, B = 7, C = -5$$

1
 $\sin$
 $\frac{1-1}{2}$
 $\sin$
 $= \frac{1}{6}$
 $= \frac{1}{6}$
 $\div$
 $2 \sin$
 $(2 \sin$
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Q4) $\int_0^1 \sin(1+x^2) dx$

$$\frac{1-0}{2} = \frac{1}{2}$$

$$\sin(1+x^2) = \frac{1}{3} \times \frac{1}{2} \left[f(0) + 4f\left(\frac{1}{2}\right) + f(1) \right]$$

$$= \frac{1}{6} \left[\sin 1 + 4 \sin 1.25 + \sin 2 \right]$$

$$= \frac{1}{6} [5.546]$$

$$\approx 0.924$$

$$2 \sin^2 x - 3 \sin x + 1 = 0$$

$$(2 \sin x - 1)(\sin x - 1) = 0$$

$$\sin x = \frac{1}{2} \quad \text{or} \quad \sin x = 1$$

$$x = \frac{\pi}{6} \text{ or } \pi - \frac{\pi}{6} \text{ or } \frac{\pi}{2}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$$

Amount owing after 1 mth

$$30000 \left(1 + \frac{1.1}{100} \right) - m$$

$$30000(1.011) - m$$

Amt. owing after 2 months

$$[30000(1.011) - m] 1.011 - m$$

$$3(x+1) + B + 030000(1.011)^2 - m(1+1.011)$$

$$30663.63 - 2.011m$$

Amt. owing after 48 payments $\times 6$

$$30000(1.011)^{48} - m(1 + 1.011 + \dots + 1.011^{47})$$

$$30000(1.011)^{48} = \frac{m \times 1(1.011^{48} - 1)}{1.011 - 1}$$

$$7119.97 \dots = 62.787m$$

$$m = 807.8060$$

Repayments are \$807.81

Question 9

a) $3x^2 + 5x - 2 = 0$

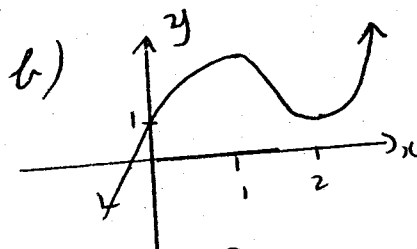
i) $\alpha + \beta = -\frac{5}{3}$

ii) $\alpha\beta = -\frac{2}{3}$

iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\beta^2 + \alpha^2}{(\alpha\beta)^2}$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$= \frac{\frac{25}{9} + \frac{4}{3}}{\frac{4}{9}} = \frac{37}{9} \times \frac{9}{4} = \frac{37}{4}$$



c) $V = \pi \int_0^2 (e^x + 1)^2 dx$

$$= \pi \int_0^2 e^{2x} + 2e^x + 1 dx$$

$$= \pi \left[\frac{e^{2x}}{2} + 2e^x + x \right]_0^2$$

$$= \pi \left[\frac{e^4}{2} + 2e^2 + 2 - \left(\frac{1}{2} + 2 \right) \right]$$

$$= \pi \left[\frac{e^4}{2} + 2e^2 - \frac{1}{2} \right] u^3$$

d) $\frac{1}{2}(5x-1) = 4 - \frac{1}{3}(x-2)$

$$3(5x-1) = 24 - 2(x-2)$$

$$15x - 3 = 24 - 2x + 4$$

$$17x = 31$$

$$x = \frac{31}{17}$$

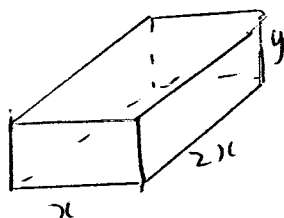
Question 10

Volume = 4

$$x \times 2x \times y = 4$$

$$2x^2 y = 4$$

$$y = \frac{4}{2x^2} = \frac{2}{x^2}$$



$$\text{Cost} = (2 \times 2x^2) \times 15 + (2 \times 2x \times y + 2 \times x \times y) \times 10$$

$$\begin{aligned} C &= 15 \times 4x^2 + 6xy \times 10 \\ &= 60x^2 + 6x \times \frac{2}{x^2} \times 10 \\ &= 60x^2 + \frac{120}{x} \end{aligned}$$

1) For cheapest cost,

$$\frac{dC}{dx} = 0 \quad \text{ie} \quad 120x - \frac{120}{x^2} = 0$$

$$120x = \frac{120}{x^2}$$

$$x^3 = 1$$

$$x = 1$$

$$\frac{d^2C}{dx^2} = 120 + \frac{240}{x^3} > 0$$

$\therefore$ Minimum cost when $x = 1$

$\therefore$ Measurements are

1m by 2m by 2m.

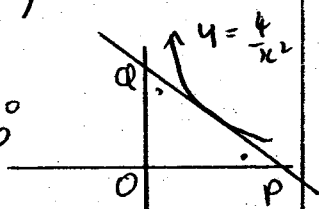
b) $\angle OPQ = \angle OQP$ (given)

$$\angle OPQ + \angle OQP + \angle QOP = 180^\circ$$

($\angle$ sum of Δ)

$$\angle OPQ = \angle OQP = 45^\circ \quad \text{as } \angle QOP = 90^\circ$$

$\therefore$ QP makes an $\angle$ of 135° with the +ve x axis so $m_{PQ} = \tan 135^\circ = -1$



$$\text{ii) } m_{\text{tangent to } y = \frac{4}{x^2} \text{ is } -1$$

$$\therefore \frac{dy}{dx} = -1 \quad \text{ie} \quad -\frac{8}{x^3} = -1$$

$$8 = x^3$$

$$x = 2$$

When $x = 2$, $y = 1$ so the tangent PQ touches the curve at $(2, 1)$

iii) Equation is $y - 1 = -1(x - 2)$ ie $x + y - 3 = 0$

$$\text{c) } y = \cot x = \frac{\cos x}{\sin x}$$

$$\frac{dy}{dx} = \frac{\sin x \times -\sin x - \cos x \times \cos x}{(\sin x)^2}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x$$